Database Systems

Lecture 13

Database Systems

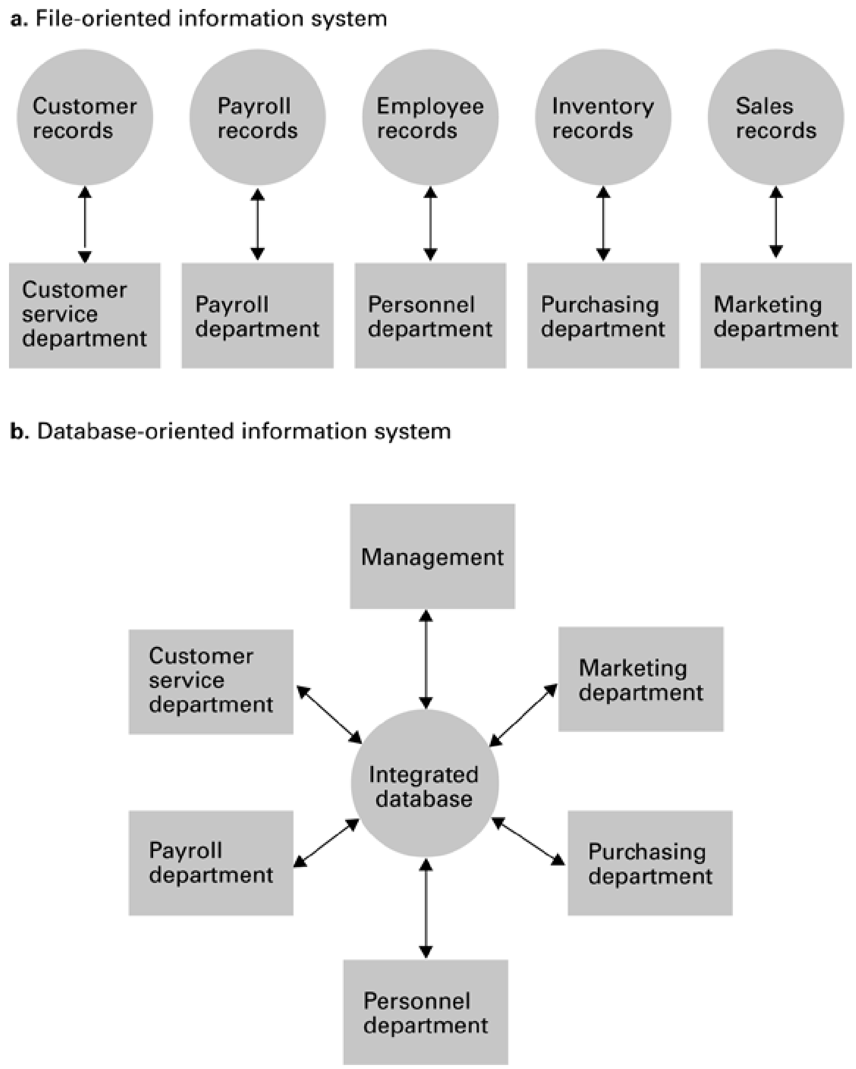
- Database Fundamentals
- The Relational Model
- Object-Oriented Databases
- Maintaining Database Integrity
- Traditional File Structures

Database

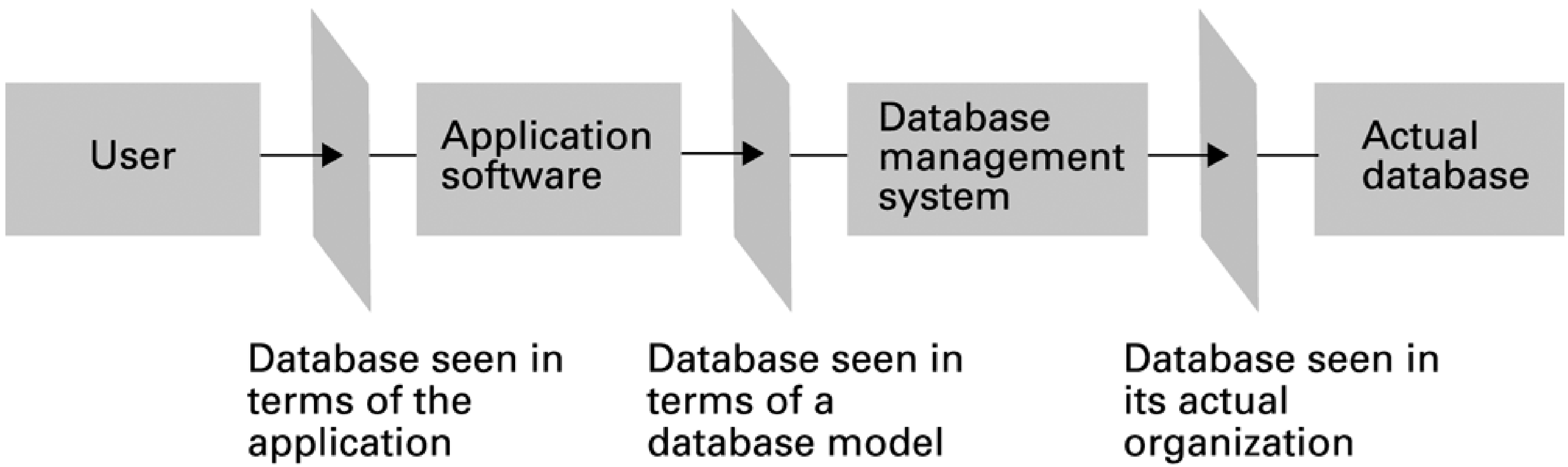
A collection of data that is multidimensional in the sense that internal links between its entries make the information accessible from a variety of perspectives

- **Database System:** Multidimensional data collection, internal links between its entries make the information accessible from a variety of perspectives
- **Flat File System:** One-dimensional file storage system that presents its information from a single point of view

A file versus a database organization



The conceptual layers of a database implementation



Schemas

- **Schema:** A description of the structure of an entire database, used by database software to maintain the database
- **Subschema:** A description of only that portion of the database pertinent to a particular user's needs, used to prevent sensitive data from being accessed by unauthorized personnel

Database Management Systems

- **Database Management System** (DBMS): A software layer that manipulates a database in response to requests from applications
- **Distributed Database:** A database stored on multiple machines
  - DBMS will mask this organizational detail from its users
- **Data independence:** The ability to change the organization of a database without changing the application software that uses it

Database Models

- **Database model:** A conceptual view of a database
  - Relational database model
  - Object-oriented database model
  - Hierarchical model
- Relational model is the most popular model
  - The database is a collection of tables
  - Each table is called a **relation**
  - Each column in the table an **attribute**
  - Each row in the table a **tuple (record)**

A relation containing employee information

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
•	•	•	•
•	•	•	•
•	•	•	•

- The schema can also be denoted as follows

```
employee_info
{
    Empl_Id: string
    Name: string
    Address: string
    SSN: int
} primary key (Empl_Id, Name, SSN)
```

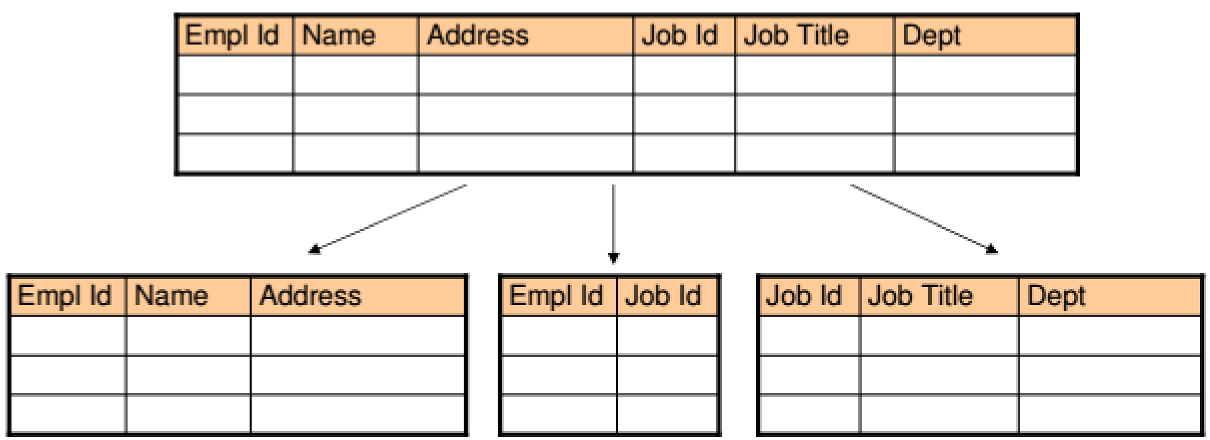
Relational Design

- Avoid multiple concepts within one relation
  - Can lead to redundant data
  - Deleting a tuple could also delete necessary but unrelated information

Empl Id	Name	Address	SSN	Job Id	Job Title	Skill Code	Dept	Start Date	Term Date
25X15	Joe E. Baker	33 Nowhere St.	111223333	F5	Floor manager	FM3	Sales	9-1-2007	9-30-2008
25X15	Joe E. Baker	33 Nowhere St.	111223333	D7	Dept. head	K2	Sales	10-1-2008	*
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999	F5	Floor manager	FM3	Sales	10-1-2007	*
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555	S25X	Secretary	T5	Personnel	3-1-1999	4-30-2006
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555	S26Z	Secretary	T6	Accounting	5-1-2006	*
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Improving a Relational Design

- **Decomposition:** Dividing the columns of a relation into two or more relations, duplicating those columns necessary to maintain relationships
  - **Lossless** or **nonloss** decomposition: A “correct” decomposition that does not lose any information



An employee database consisting of three relations

EMPLOYEE relation

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555

JOB relation

Job Id	Job Title	Skill Code	Dept
S25X	Secretary	T5	Personnel
S26Z	Secretary	T6	Accounting
F5	Floor manager	FM3	Sales
•	•	•	•
•	•	•	•
•	•	•	•

ASSIGNMENT relation

Empl Id	Job Id	Start Date	Term Date
23Y34	S25X	3-1-1999	4-30-2006
34Y70	F5	10-1-2007	*
23Y34	S26Z	5-1-2006	*
•	•	•	•
•	•	•	•
•	•	•	•

Finding the departments in which employee 23Y34 has worked

EMPLOYEE relation

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
⋮	⋮	⋮	⋮

JOB relation

Job Id	Job Title	Skill Code	Dept
S25X	Secretary	T5	Personnel
S26Z	Secretary	T6	Accounting
F5	Floor manager	FM3	Sales
⋮	⋮	⋮	⋮

are contained in the personnel and accounting departments.

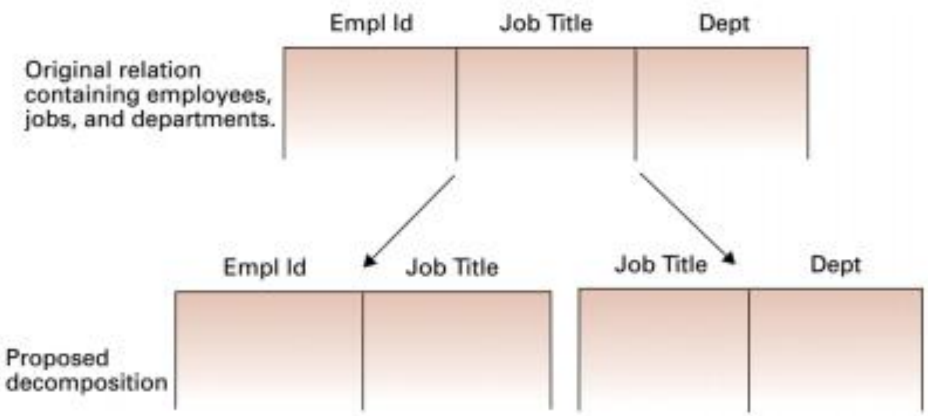
ASSIGNMENT relation

Empl Id	Job Id	Start Date	Term Date
23Y34	S25X	3-1-1999	4-30-2006
34Y70	F5	10-1-2007	*
23Y34	S26Z	5-1-2006	*
⋮	⋮	⋮	⋮

The jobs held by employee 23Y34

Lossless Decomposition

- Decompositions can cause losses
- For example, if below relations is decomposed as follows, the information about the coupling of employee-department is lost.
- We cannot use JobTitle as a field to match employee and department, because many departments can hire people with the same JobTitle.
- For example, SOCIE and ICE departments at INHA hire Professors.



Relational Operations

- **Select:** Choose rows
- **Project:** Choose columns
- **Join:** Assemble information from two or more relations

The SELECT operation

EMPLOYEE relation

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
⋮	⋮	⋮	⋮

NEW ← SELECT from EMPLOYEE where EmplId = “34Y70”

NEW relation

Empl Id	Name	Address	SSN
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999

The PROJECT operation

EMPLOYEE relation

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
24Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
•	•	•	•
•	•	•	•
•	•	•	•

↓

MAIL ← PROJECT Name, Address from EMPLOYEE

↓

MAIL relation

Name	Address
Joe E. Baker	33 Nowhere St.
Cheryl H. Clark	563 Downtown Ave.
G. Jerry Smith	1555 Circle Dr.
•	•
•	•
•	•

The JOIN operation

Relation A

V	W
r	2
t	4
p	6

Relation B

X	Y	Z
5	g	p
4	d	e
2	m	q
4	t	f

C ← JOIN A and B where A.W = B.X

Relation C

A.V	A.W	B.X	B.Y	B.Z
r	2	2	m	q
t	4	4	d	e
t	4	4	t	f

Another example of the JOIN operation

Relation A

V	W
r	2
t	4
p	6

Relation B

X	Y	Z
1	g	p
4	d	e
2	m	q
5	t	f

C ← JOIN A and B where A.W < B.X

Relation C

A.V	A.W	B.X	B.Y	B.Z
r	2	4	d	e
r	2	5	t	f
t	4	5	t	f

An application of the JOIN operation

ASSIGNMENT relation

Empl Id	Job Id	Start Date	Term Date
23Y34	S25X	3-1-1999	4-30-2006
34Y70	F5	10-1-2007	*
25X15	S26Z	5-1-2006	*
•	•	•	•
•	•	•	•
•	•	•	•

JOB relation

Job Id	JobTitle	Skill Code	Dept
S25X	Secretary	T5	Personnel
S26Z	Secretary	T6	Accounting
F5	Floor manager	FM3	Sales
•	•	•	•
•	•	•	•
•	•	•	•

NEW1 ← JOIN ASSIGNMENT and JOB where ASSIGNMENT. JobId = JOB.JobId

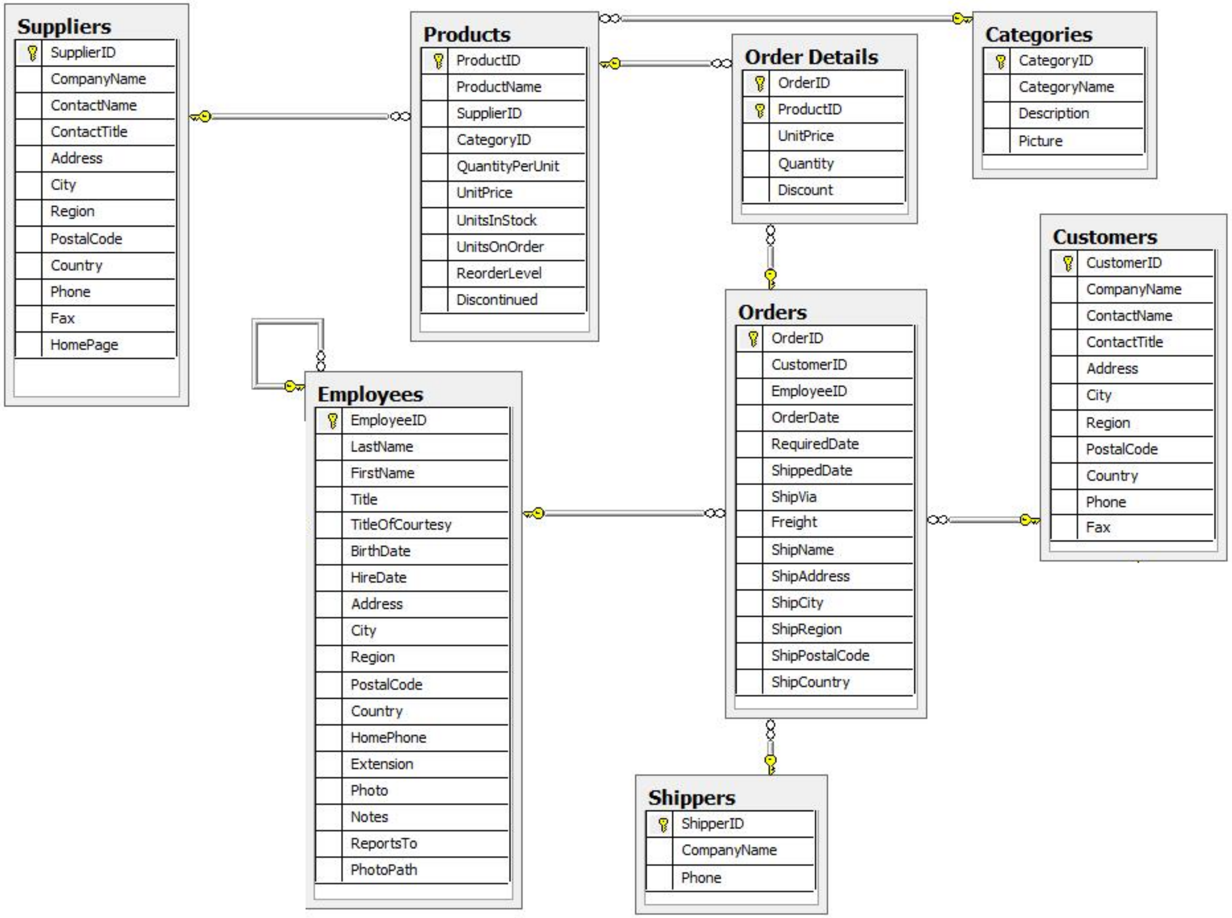
NEW1 relation

ASSIGNMENT Empl Id	ASSIGNMENT Job Id	ASSIGNMENT StartDate	ASSIGNMENT TermDate	JOB Job Id	JOB JobTitle	JOB SkillCode	JOB Dept
23Y34	S25X	3-1-1999	4-30-2006	S25X	Secretary	T5	Personnel
34Y70	F5	10-1-2007	*	F5	Floor manager	FM3	Sales
25X15	S26Z	5-1-2006	*	S26Z	Secretary	T6	Accounting
•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•
•	•	•	•	•	•	•	•

Structured Query Language (SQL)

- SQL is the most popular language used to create, modify, retrieve ad manipulate information from relational DBMS
- SQL was originally designed by IBM in 1970s, and became an ISO standard in 1987
- In SQL, there are some operations to manipulate tuples
  - insert
  - update
  - delete
  - select

Database Schema



SQL Examples

```
SELECT CustomerName, Phone, City
FROM Customers;

SELECT o.OrderID, o.OrderDate, c.CustomerName, c.City,
c.PostalCode, c.Country FROM Orders o, Customers c WHERE
o.CustomerID=c.CustomerID


SELECT c.CustomerName, p.ProductName,
       o.OrderID, o.OrderDate,
       p.Unit, od.Quantity,
       p.Price
FROM Customers c, Orders o, OrderDetails od, Products p
WHERE      c.CustomerID=o.CustomerID
       AND o.OrderID=od.OrderID
       AND p.ProductID=od.ProductID
       AND p.ProductName='Tofu'
```

SQL Examples (continued)

- ```
  INSERT INTO Employees(LastName,
  FirstName, BirthDate)
  VALUES (‘Abdullaev', ‘Sarvar', '1968-
  12-08');
  ```

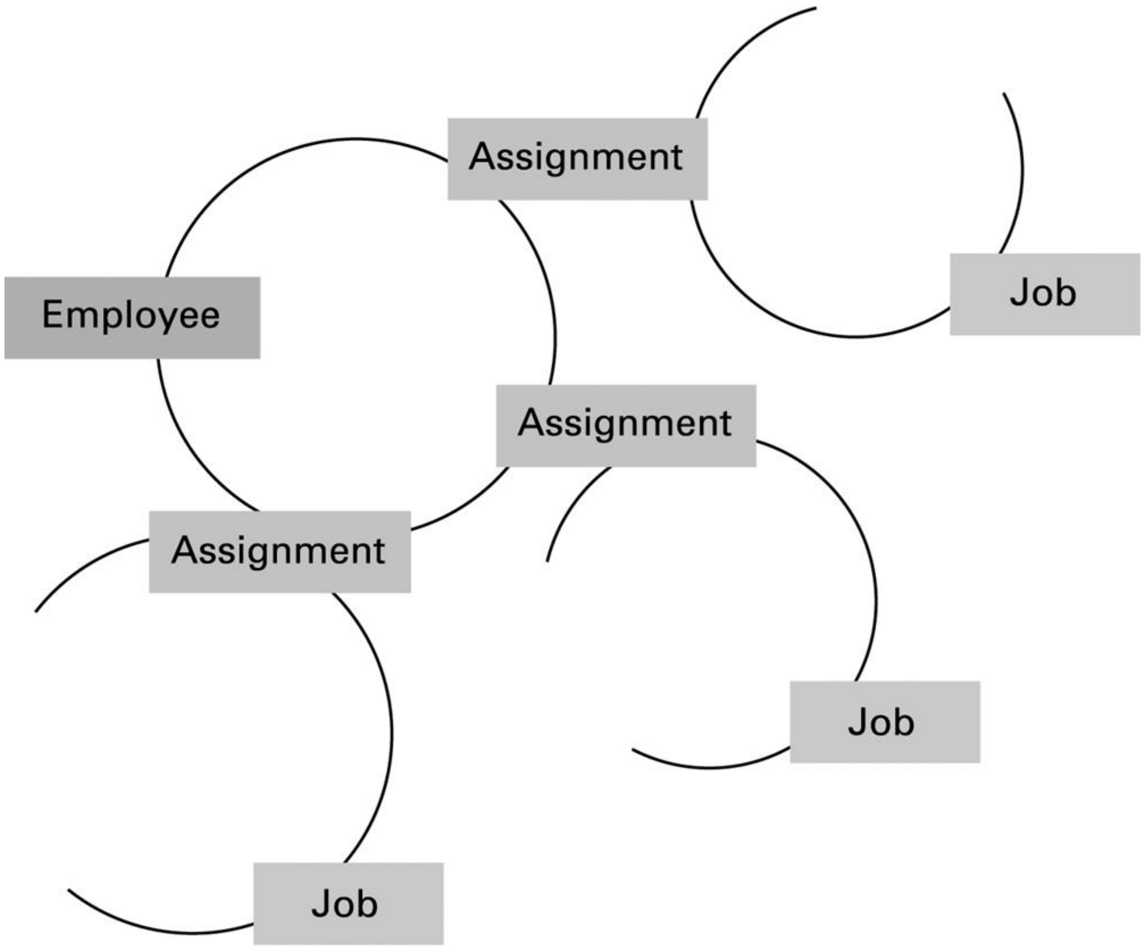
- ```
  DELETE FROM Employees
  WHERE LastName = ‘Abdullaev';
  ```

- ```
  UPDATE Employee
  SET BirthDate = ‘1985-01-01'
  WHERE Name = ‘Abdullaev';
  ```
  ```

Object-oriented Databases

- **Object-oriented Database:** A database constructed by applying the object-oriented paradigm
  - Each entity stored as a ***persistent*** object
  - Relationships indicated by links between objects
  - DBMS maintains inter-object links
- Advantages of OO Databases:
  - Exotic data types such as audio, video, graphics, data structures and custom formats
  - Seamless integration with OOP languages
  - Good for caching
- Some examples of OO Databases:
  - MongoDB, ObjectDB, GemStone, Objectivity/DB

The associations between objects in an object-oriented database



Maintaining Database Integrity

- **Transaction:** A sequence of operations that must all happen together
  - Example: transferring money between bank accounts
- **Transaction log:** A non-volatile record of each transaction's activities, built before the transaction is allowed to execute
  - **Commit point:** The point at which a transaction has been recorded in the log
  - **Roll-back:** The process of undoing a transaction

Simultaneous Access Problems

Temporary Update Problem

T1	T2
read_item(X) X = X - N write_item(X)	
	read_item(X) X = X + M write_item(X)
read_item(Y)	

Incorrect Summary Problem

T1	T2
	sum = 0 read_item(A) sum = sum + A
read_item(X) X = X - N write_item(X)	
	read_item(X) sum = sum + X read_item(Y) sum = sum + Y
read_item(Y) Y = Y + N write_item(Y)	

Simultaneous Access Problems

Lost Update Problem

T1	T2
read_item(X) X = X + N	
	X = X + 10 write_item(X)

Phantom Read Problem

T1	T2
Read(X)	
	Read(X)
Delete(X)	
	Read(X)

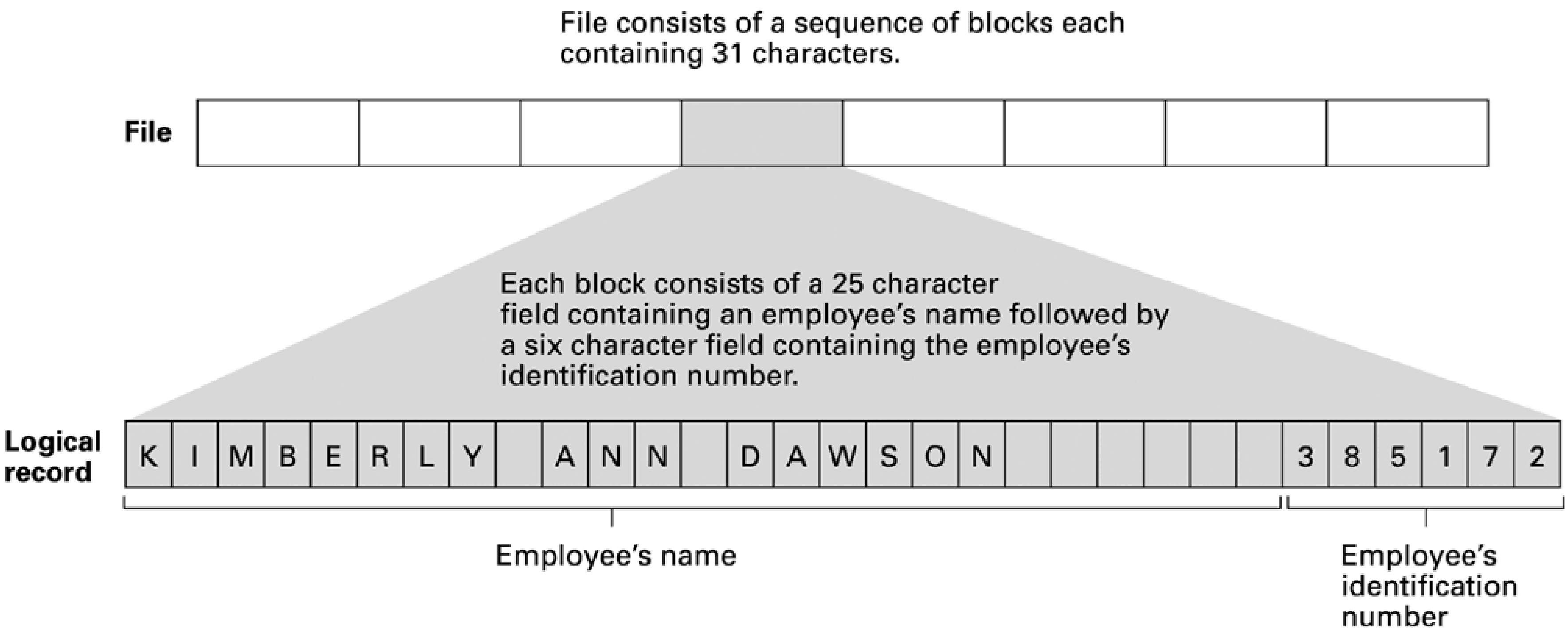
Maintaining database integrity (continued)

- **Locking** = preventing others from accessing data being used by a transaction
  - **Shared** lock: used when reading data
  - **Exclusive** lock: used when altering data
- A common way to resolve deadlock in DBMS is the ***wound-wait protocol***:
  - In a hold-and-wait deadlock situation, the data item held by the younger transaction will be forcibly retrieved by the older transaction.

Sequential Files

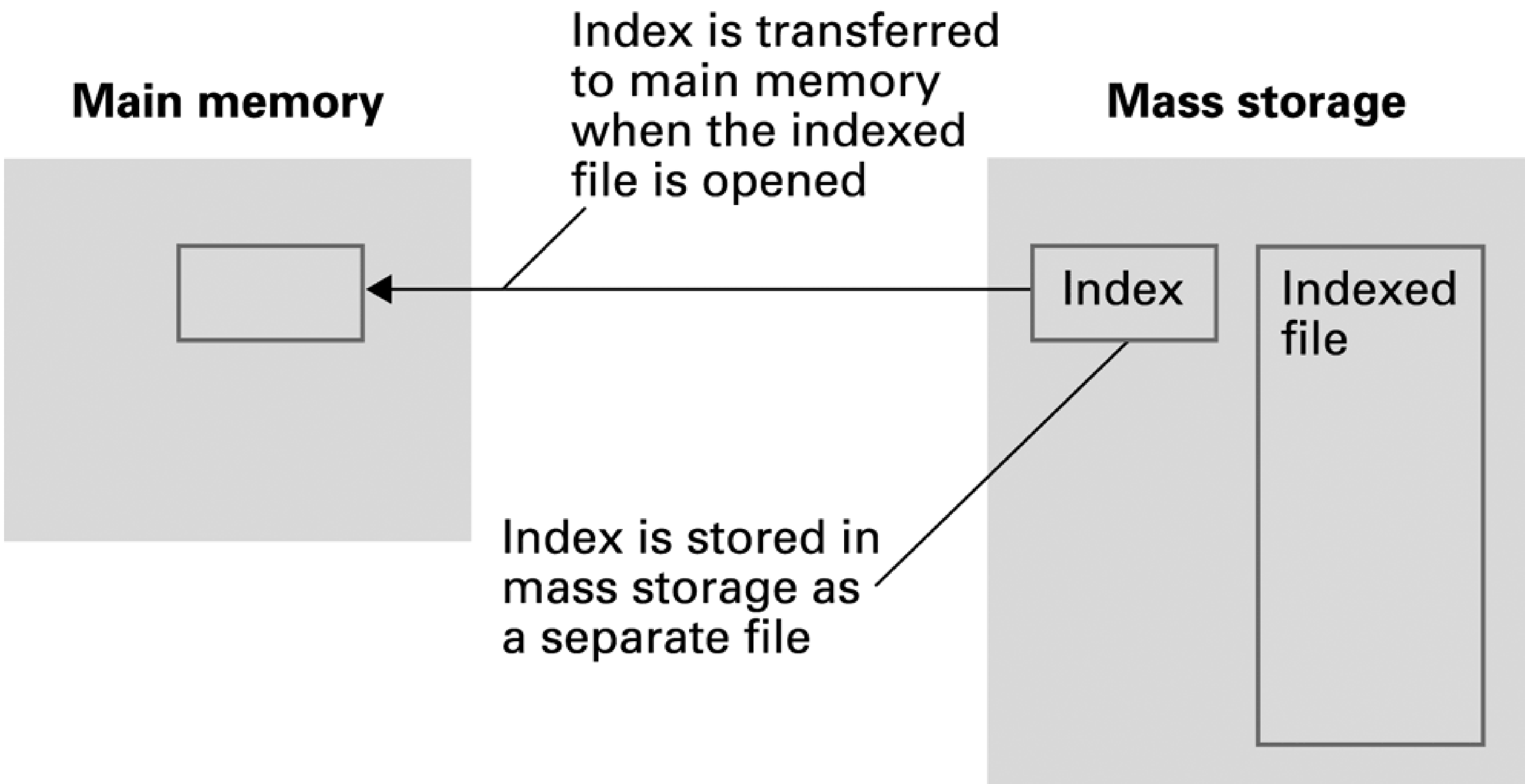
- **Sequential file:** A file whose contents can only be read in order
  - Reader must be able to detect end-of-file (EOF)
  - Data can be stored in logical records, sorted by a key field
    - Greatly increases the speed of batch updates

The structure of a simple employee file implemented as a text file



Indexed Files

- **Index:** A list of key values and the location of their associated records



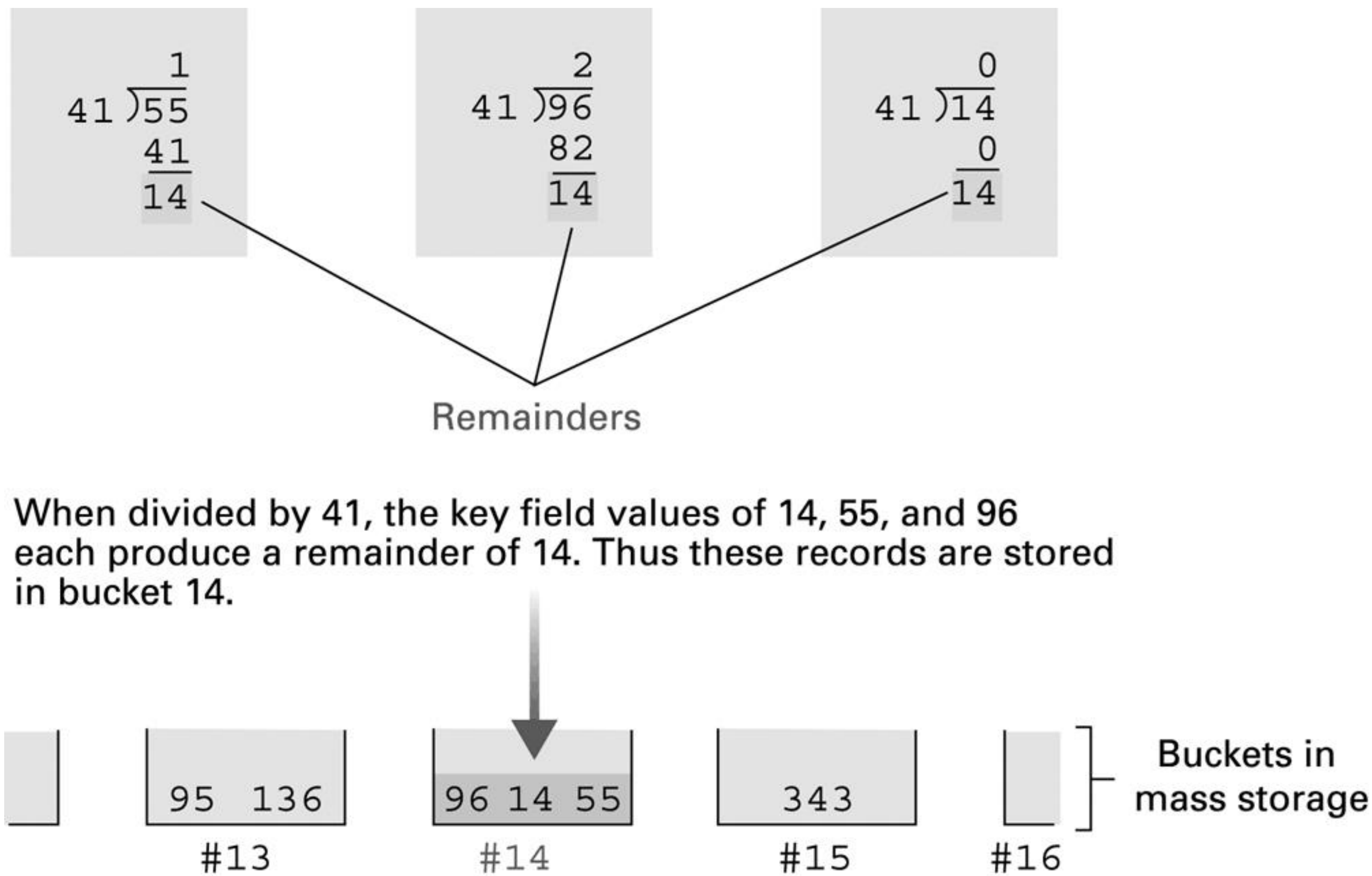
Hashing

- Each record has a key field
- The storage space is divided into **buckets**
- A **hash function** computes a bucket number for each key value
- Each record is stored in the bucket corresponding to the hash of its key

Hashing the key field value 25X3Z to one of 41 buckets

Key field value:	25X3Z
ASCII representation:	0011001000110101010110000011001101011010
Equivalent base ten value:	215,643,337,562
Remainder after division by 41:	3
Bucket number:	3

The rudiments of a hashing system



Collisions in Hashing

- **Collision:** The case of two keys hashing to the same bucket
  - Major problem when table is over 75% full
  - Solution: increase number of buckets and rehash all data

Data Mining

- **Data Mining:** The area of computer science that deals with discovering patterns in collections of data
- **Data warehouse:** A static data collection to be mined
  - **Data cube:** Data presented from many perspectives to enable mining

Data Mining Strategies

- Class description
- Class discrimination
- Cluster analysis
- Association analysis
- Outlier analysis
- Sequential pattern analysis

Social Impact of Database Technology

- Problems
  - Massive amounts of personal data are being collected
    - Often without knowledge or meaningful consent of affected people
  - Data merging produces new, more invasive information
  - Errors are widely disseminated and hard to correct
- Remedies
  - Existing legal remedies often difficult to apply
  - Negative publicity may be more effective

Suggested Activity/Reading

- Learn SQL:
  - http://www.w3schools.com/sql
- Read the remaining chapters of the textbook:
  - Chapter 10: Computer Graphics
  - Chapter 11: Artificial Intelligence
  - Chapter 12: Theory of Computation